Peichun Hua

Computer Science and Engineering, School of Data Science
The Chinese University of Hong Kong, Shenzhen
Github — Website — Google Scholar — peichunhua@link.cuhk.edu.cn

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EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) — September 2022 — June 2026
B.Eng in Computer Science and Engineering, School of Data Science (SDS)
First Class Honors, Outstanding Graduate
Overall GPA: 3.87/4.00 — Major GPA: 4.00/4.00 — Rank: 2/132

University of North Carolina, Chapel Hill — September 2024 - May 2025
Overall GPA: 4.00/4.00
Exchange Student — Computer Science Department

RESEARCH INTERESTS

- Responsible AI and Adversarial Machine Learning; [14], [8], [4], [2]
- Information Retrieval: retrieval-augmented generation (RAG), agentic RAG, their algorithms, applications, and infrastructure; [10], [7], [6], [5], [3], [1]
- AI and IoT Security: secure hardware design; ML model IP protection; [16], [15], [13], [9]
- Privacy-Enhancing Technologies: the application of differential privacy (DP), homomorphic encryption (HE), secure multi-party computation (MPC), especially in AI systems; [7], [6]

PUBLICATION

* denotes co-first authors of equal contribution.

Preprint / Under Peer Review

[1] **Peichun Hua**, He Sun, Zirui Zhuang, Qi Qi, Yiming Qiu, Shinan Liu, Yunming Xiao. "Tessera: Structure-Completed Evidence Units for Web Retrieval-Augmented Generation." *Under Review.* (Paper (arXiv), Code).

[2] Junguang Yao, **Peichun Hua**, Hanxiu Zhang, Yue Zheng. "Label-Budgeted Attribute Inference from Protected Features in Edge-Cloud Split Multimodal LLMs." *Under Review.* (Paper (arXiv)).

[3] Suting Chen*, **Peichun Hua***, Yunming Xiao. "Relevance Is Not Sufficient Evidence: Detecting Evidence Gaps Before Generation in RAG." *Under Review.* (Paper (arXiv)).

[4] Beining Xu*, **Peichun Hua***, Yunming Xiao. "Backdoor in the Loop: Compromising Agentic Search via Malicious Retrievers." *Under Review.* (Paper (arXiv)).

[5] **Peichun Hua**, Yunming Xiao. "Matryoshka Hash Representations for Model-Aware Compact Semantic Retrieval." *Under Review.* (Paper (arXiv)).

[6] **Peichun Hua**, Yunming Xiao. "SPRUCE: Scalable Private Outsourced Retrieval Using Compact Embeddings." *Under Review.* (Paper (arXiv))

[7] **Peichun Hua**, Danyang Chen, Junan Zhang, Haifeng Sun, Jingyu Wang, Diwen Xue, Mingyu Li, Yunming Xiao. "Pointing the Way, Hiding the Destination: Practical Private Dense Retrieval at Scale." *Under Review* (Paper (arXiv)).

[8] **Peichun Hua**, Haoxuan Xu, Mengyuan Li. "Behavioral Skill Reconstruction: Reconstructing Hidden Functionality from LLM Agent Skills." *Under Review* (Paper (arXiv)).

[9] **Peichun Hua**, Yue Zheng. "ChaosFormer: Hardware-Bound Edge Transformer Model Obfuscation for Intellectual Property Protection." *Under Major Revision, ACM Transactions on Design Automation of Electronic Systems (TODAES)*, PDF).

Published / To Appear

[10] **Peichun Hua**, Yunming Xiao. "Semantics Delivery Network: Rethinking Web Retrieval Infrastructure for LLM Agents." To appear at the 25th ACM Workshop on Hot Topics in Networks (HotNets 2026) (AR = 43/188 ≈ 22.9%, Code, Paper, PDF)

[11] Junguang Yao, **Peichun Hua**, Yue Zheng. "Enroll Once, Identify Everywhere: Few-Shot Cross-Scenario Visual Speaker Recognition via Event-driven SlowFast Networks." To appear at the Pacific Rim International Conference on Artificial Intelligence (PRICAI 2026) (Oral Presentation, CORE: B, AR = 542/1825 ≈ 29.7%, Code, PDF).

[12] Danyang Chen, Yufeng Gu, Yibo Huang, Chengxuan Pei, **Peichun Hua**, Yang Zhou, Yunming Xiao. "Unlocking Software-defined GPU Fabric Scheduling in the LLM Era." In the 17th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys 2026) (AR = 19/43 ≈ 44.2%, Code, Paper, PDF, Slides).

[13] Xin Wang, **Peichun Hua**, Chip Hong Chang, Wenye Liu, Yue Zheng. "A Unified Open-Set Framework for Scalable PUF-Based Authentication of Heterogeneous IoT Devices." To appear at the IEEE International Conference on Systems, Man, and Cybernetics (SMC 2026) (CORE: B, Code, Paper, PDF).

[14] **Peichun Hua**, Hao Li, Shanghao Shi, Zhiyuan Yu, Ning Zhang. "Rethinking Jailbreak Detection of Large Vision Language Models with Representational Contrastive Scoring." In the Annual Meeting of the Association for Computational Linguistics (ACL 2026). (Main Conference, **Oral Presentation**, CORE: A*, AR = 2296/12148 ≈ 18.9%, Code, Paper, PDF, Presentation, Slides).

[15] **Peichun Hua**, Hanxiu Zhang, Tuo Li, Yue Zheng, Wenye Liu. "Live Demonstration: Hardware-Bound IP Protection for OWL-ViT on Edge Devices." In the IEEE International Symposium on Circuits and Systems (ISCAS 2026) (Code, Paper, PDF).

[16] **Peichun Hua**, Hanxiu Zhang, Tuo Li, Yue Zheng. "Securing On-device Transformer with Hardware Binding and Reversible Obfuscation." In the Annual Computer Security Applications Conference (ACSAC 2025) (CORE: A, AR = 84/446 ≈ 18.8%, Code, Paper, PDF, Presentation, Slides).

AWARDS

Academic Honors

- **Outstanding Graduate**, School of Data Science, CUHK-Shenzhen 2026
- Dean's List, School of Data Science, CUHK-Shenzhen 2022–2023, 2023–2024, 2025-2026
- Academic Performance Scholarship (Class B, Top <2%), School of Data Science 2023–2024
- Academic Performance Scholarship (Class C, Top <5%), School of Data Science 2022–2023
- University Research Award (with scholarship) Fall 2024, Spring 2025

Research Awards/Grants

- *Student Travel Grant*, 25th ACM Workshop on Hot Topics in Networks, HotNets 2026
- *Student Conferenceship (Travel Grant)*, 41st Annual Computer Security Applications Conference, ACSAC 2025
- *Artifact Available*, *Artifact Reviewed*, and *Artifact Reproducible* Badges — IEEE, for the paper "Securing On-device Transformer with Hardware Binding and Reversible Obfuscation" at ACSAC 2025
- McKelvey International Student Research Internship Program, Washington University in St. Louis Summer 2025

EXPERIENCE

University of Southern California Los Angeles, CA, USA
Research Assistant (Remote) May 2026 – August 2026

- Advisor: Prof. Mengyuan Li
- Topic: Agentic AI Security. Work completed independently and is under review at ACL Rolling Review (ARR).

School of Data Science, CUHK-Shenzhen Shenzhen, China
Research Assistant September 2025 – April 2026 (Part-time); April 2026 - Present (Full-time)

- Advisor: Prof. Yunming Xiao
- Topics: 1) The security and privacy of RAG pipeline; 2) The application of deep hashing in information retrieval; 3) The infrastructure and performance of RAG in the real web.
- Five papers completed independently. Two full-length papers under review at top system/security conferences, one full-length paper at a top AI conference, and one vision paper accepted by ACM HotNets.

McKelvey School of Engineering, Washington University in St. Louis Missouri, MO, USA
Summer Research Intern May 2025 – September 2025

- Fully funded by International Student Research Internship Program
- Advisor: Prof. Ning Zhang
- Topic: Multi-modal Jailbreak Detection with interpretability methods instantiated with statistical tests and contrastive learning. Achieve highly effective, generalizable, and lightweight attack detection.
- Work completed independently. **Paper accepted by ACL 2026 Oral.**

School of Science and Engineering, CUHK-Shenzhen Shenzhen, China
Undergraduate Research Intern June 2024 – May 2025, October 2025 - May 2026

- Advisor: Prof. Yue Zheng
- Topic: IoT Security, IP (intellectual property) Protection of on-device transformer models.
- Conferred University Research Award (URA, with scholarship) twice. Work completed independently and accepted by ACSAC 2025, IEEE ISCAS 2026, and the extended journal version under major revision at ACM TODAES. One co-first author paper under review at a major security conference.

SKILLS

- **Coursework:** Achieved A grade in all listed *graduate-level* coursework
  Cryptography | 3D Computer Vision | Hardware Security and Side-Channels
  Efficient Deep Learning | Research Topics in Computer Security | Formal Methods in Computer Security
- **Language Proficiency:** TOEFL 108 (R29+L29+S23+W27), GRE 328 (V161+Q167+3.5)

SERVICES

- Reviewer, IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP), 2026
- Reviewer, ACL Rolling Review, 2026
- Reviewer, The Pacific Rim International Conference on Artificial Intelligence (PRICAI), 2026
- Reviewer, International Conference on Learning Representations (ICLR), 2027
- Reviewer, ACM The Web Conference (WWW), 2027